

PROBLEM STATEMENT · TRACK 6 - OPEN INNOVATION

AI Face Recognition Attendance System

Powered by FaceNet · CCTV Monitoring · Real-Time Dashboard

TRACK

6 - Open Innovation

PS ID

6

TEAM NAME

Telos

TEAM LEADER

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What is the Solution?

- CCTV monitors classroom — AI auto-marks attendance in real time
- FaceNet generates 512D face embeddings for each registered student
- Frames extracted from live feed every 5 seconds for processing
- Multiple-frame verification prevents false positives
- Live Streamlit dashboard + automatic CSV / Excel report export

How It Works (Summary)

- Student registered once with passport photo + augmented variants
- CCTV feed → OpenCV extracts frame → FaceNet embeds each face
- Cosine similarity match → 'already-marked' list prevents re-marking
- Attendance saved with timestamp → visible on dashboard instantly

Problem It Solves

- Manual roll calls waste 5–10 min per class
- Proxy attendance via paper goes completely undetected
- No way to track live classroom occupancy
- Attendance records get lost or tampered easily

Innovation & Uniqueness

- FaceNet 512D embeddings → higher accuracy than older libs
- CCTV-based — zero student interaction required
- Multi-face: marks entire class in one single frame
- Anti-spoofing layer: rejects printed photos / video replays
- Fully offline — data never leaves campus server
- Role-based portal: Admin / Teacher / Student views

Python 3.10

OpenCV

FaceNet

MTCNN

TensorFlow

SQLite/CSV

Streamlit

NumPy/Pandas

Anti-Spoofing

01 Student Registration

- Passport photo + 5 augmented variants collected
- FaceNet generates 512D embedding per photo
- Average embeddings stored in SQLite DB

02 Classroom Monitoring

- CCTV stream connected via RTSP / USB
- OpenCV extracts 1 frame every 5 seconds
- Only frames with detected faces are processed

03 Face Processing

- MTCNN detects all faces in each frame
- Each face cropped and aligned to 160x160 px
- FaceNet extracts 512D embedding per face

04 Matching

- Cosine similarity vs all stored embeddings
- Threshold ≥ 0.80 confirms a match
- Unrecognised faces flagged for review

05 Attendance Logic

- Maintains 'already-marked' list per session
- Student marked present only once per class
- 3-frame consensus needed to confirm presence

06 Final Output

- Record saved: name + date + timestamp
- Dashboard updates in real time automatically
- Auto CSV / Excel export when class ends

Rs. 0–500

Total Setup Cost

100%

Open-Source Stack

< 1 Day

Deployment Time

5 sec

Frame Interval

Practical Feasibility

- Existing college CCTV — no new hardware needed
- Runs on any mid-range PC or server (Python 3.10+)
- One-time student registration takes under 20 minutes
- Works fully offline — no internet dependency
- Scales from 30 to 1000+ students with minor config
- CCTV integration via standard RTSP protocol

Challenges & How We Solve Them

Poor lighting / face occlusion

Histogram equalisation + MTCNN robust detection

Students wearing masks

Retrain with masked-face dataset variants

Large class causes processing lag

Frame skip every 5 sec + multi-thread pipeline

Privacy of biometric data

Only encrypted 512D vectors stored — no raw photos

Embedding drift over semesters

Scheduled re-registration reminder each semester

Impact & Benefits

Who Benefits · How Much · Why It Matters

5-10 min

Saved Per Lecture

~80%

Admin Effort Cut

0 Proxy

Possible

95%+

FaceNet Accuracy

Rs. 0

Recurring Cost

For Students

- No proxy — CCTV sees who is really present
- Instant attendance confirmation on portal
- View personal attendance % in real time
- Fairer and completely transparent system
- Zero effort — just walk into class

For Teachers & Admin

- Zero manual roll-call, ever
- Auto daily report emailed or exported
- Defaulter list generated automatically
- Single dashboard for all subjects
- Historical trend analytics built in

Society & Institution

- Eliminates paper waste completely
- Scales to any institution globally
- Offline — biometric data stays on campus
- Cuts admin operational costs by ~80%
- Blueprint for a truly smart campus

◆ Real-Time CCTV Processing ◆ Anti-Spoofing Built-In

❖ Live Dashboard

✦ Fully Offline

❖ Auto CSV Export

◆ FaceNet 512D Embeddings

Research FaceNet (Google Research, 2015)

arxiv.org/abs/1503.03832

Library MTCNN — Face Detection

github.com/ipazc/mtcnn

Library **OpenCV Documentation**

docs.opencv.org/4.x/

Framework TensorFlow / Keras

tensorflow.org/api_docs

Framework **Streamlit Official Docs**

docs.streamlit.io

Docs Python SQLite3 Reference

docs.python.org/3/library/sqlite3.html

Dataset LFW Dataset (UMass)

vis-www.cs.umass.edu/lfw/

Research IJSER — Real-Time Face Attendance

ijser.org — Vol.10, Issue 4, 2019

Library Silent-Face Anti-Spoofing

github.com/minivision-ai/Silent-Face-Anti-Spoofing

Library NumPy & Pandas

numpy.org | pandas.pydata.org